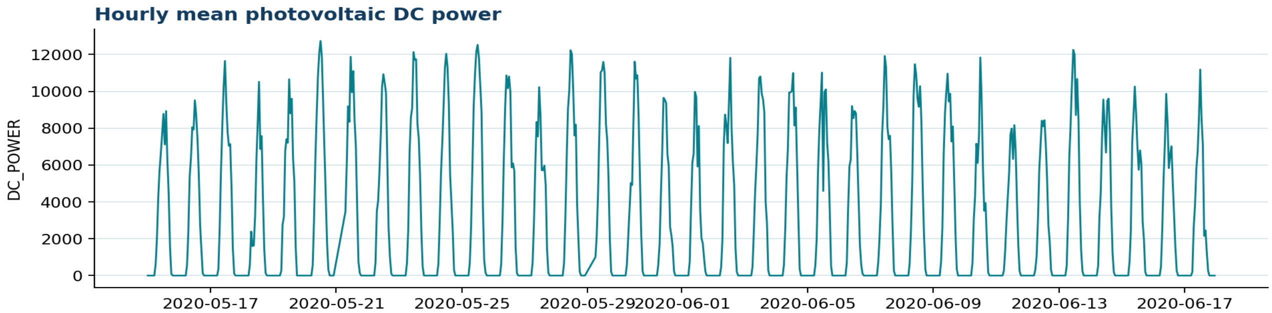
Use Case 3 - Solar-Power Forecasting

Two-hour-ahead photovoltaic generation forecasting from Plant 1 data

Dataset identity and quality

Checked CSV	use_case_3_solar_power_hourly.csv
Source	Kaggle - Solar Power Generation Data
Rows / target	796 observed timestamps / DC_POWER
Covered period	2020-05-15 00:00 to 2020-06-17 23:00
Cadence	Hourly observed aggregates
Complete-index check	816 expected hours; 20 absent timestamps
Range / zeros	0.00 to 12738.07; 358 zero observations
Missing cells / duplicates	0 missing values; 0 duplicate timestamps



Use-case role

The original collection contains inverter-level generation and plant-level weather-sensor data for two photovoltaic plants. This use case uses Plant 1 generation records only. The target is the arithmetic hourly mean of available inverter-level DC_POWER records; it is not total plant energy and not grid-injected AC power.

Preprocessing

The original measurements are primarily 15-minute observations. The supplied subset contains 796 observed hourly means. Completing the hourly index yields 816 timestamps and reveals 20 missing hours. Physical night-time zeros must be retained. The thesis workflow assigns zero to consistently non-producing night hours, interpolates short daytime gaps, and then uses forward/backward propagation for remaining gaps.

Train/test and models

After temporal-feature construction and removal of incomplete histories or targets, the thesis reports 794 forecast origins: 635 for training and tuning and 159 for chronological testing. The outputs are t+1 and t+2 hours. XGBoost uses a flattened representation with current value, lags 1, 2, 3, and 20, plus three-hour rolling mean and standard deviation. Prophet consumes the timestamped target directly.

Interpretation boundary

Two-sided interpolation and backward filling may use information after a forecast origin. Results based on this preparation are retrospective positive-path validation, not leakage-free online forecasting evidence. Deployment evaluation should use causal preprocessing fitted independently inside each training fold.